

RCIM Model-Bank Reproduction SVM Final Closure Campaign Results Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report closes the dedicated `svr` final-closure campaign prepared in:

- `doc/reports/campaign_plans/track_1/svm/2026-04-14-20-50-01_track1_svm_final_closure_campaign_plan_report.md`

The campaign executed `12` exact-paper validation runs through the dedicated launcher:

- completed runs: `12`
- failed runs: `0`
- execution window: `2026-04-14 21:09:00+02:00` to `2026-04-14 21:10:39+02:00`
- campaign artifact root:
`output/training_campaigns/track1/svm/track1_svm_final_closure_campaign_2026_04_14_20_50_01/` The campaign goal was explicit:
 - close the last residual yellow `svm` cells after the broader repair wave;
 - focus only on amplitude harmonics `40` , `156` , `240` ;
 - focus only on phase harmonic `162` ;
 - merge the best per-cell improvements back into the canonical `svm` row.

Objective And Outcome

The campaign had three concrete goals

1. fully close amplitude harmonic `156` ;
2. try to close amplitude `40` , amplitude `240` , and phase `162` ;
3. convert the `svm` row from near-closure into a practically closed row.

Outcome:

- all `12` runs completed successfully;
- all `12` runs exported ONNX artifacts with `0` failed exports;
- the campaign closed amplitude harmonic `156` on both Table `2` and Table `3` ;
- the campaign improved phase `162` , but did not fully close it;
- the `svm` row is now extremely close to full closure, with only `5` yellow cells left across Tables `2-5` . This means the campaign succeeded partially and usefully:
- it did not finish the full `svm` row;
- but it removed the strongest remaining amplitude blocker and tightened the row further without regressions.

Ranking Policy

This campaign is not a full-matrix family comparison. Each run only covers one scoped residual slice of the `SVM` row.

The serialized campaign policy is therefore

- primary metric: `cell_closure_score_desc`
- first tie breaker: `met_paper_cell_count_desc`
- second tie breaker: `open_paper_cell_count_asc`
- third tie breaker: `mean_normalized_gap_ratio_asc`
- fourth tie breaker: `max_normalized_gap_ratio_asc`
- fifth tie breaker: `improved_cell_count_desc`
- sixth tie breaker: `status_upgrade_score_desc`
- seventh tie breaker: `run_name`

Where:

- $\text{cell_closure_score} = (\text{met} + 0.5 * \text{near}) / \text{paper_cell_count}$
 - `met` means the scoped repository cell reached or beat the paper cell
 - `near` means the positive gap stayed within 25% of the paper cell
 - `status_upgrade_score` rewards `red -> yellow`, `yellow -> green`, and `red -> green` transitions versus the pre-final-closure `SVM` baseline row
- Interpretation:
- this policy exists only to serialize the required campaign-best artifacts;
 - the real scientific interpretation remains the merged post-final-closure `SVM` row across Tables 2-5 .

Campaign Ranking

Ranked Completed Runs

Rank	Run	Scope	Cells	Met	Near	Open	ClosureScore	Improved	Upgrades
1	track1_svm_amplitude_full_closure_split15	amplitudes_only	6	2	3	1	0.583	3	2
2	track1_svm_amplitude_final_closure_seed23	amplitudes_only	6	1	5	0	0.583	0	0
3	track1_svm_phase_final_closure_split15	phases_only	2	0	2	0	0.500	0	0
4	track1_svm_amplitude_40_bridge	amplitudes_only	2	0	2	0	0.500	0	0
5	track1_svm_phase_final_closure_seed11	phases_only	2	0	2	0	0.500	0	0
6	track1_svm_phase_final_closure_baseline	phases_only	2	0	2	0	0.500	0	0
7	track1_svm_phase_final_closure_split25	phases_only	2	0	2	0	0.500	0	0
8	track1_svm_phase_final_closure_seed23	phases_only	2	0	2	0	0.500	0	0
9	track1_svm_amplitude_final_closure_baseline	amplitudes_only	6	0	5	1	0.417	1	0
10	track1_svm_amplitude_hard_tail_focus	amplitudes_only	4	0	3	1	0.375	1	0
11	track1_svm_amplitude_final_closure_seed11	amplitudes_only	6	0	2	4	0.167	0	0
12	track1_svm_amplitude_hard_tail_seed11	amplitudes_only	4	0	0	4	0.000	0	0

Export Surface

Completed Runs	Exported ONNX Files	Failed Exports	Surrogate Exports
12	22	0	5

All 5 surrogate exports came from amplitude-side svr targets that still needed the existing constant-linear-regression safeguard.

Campaign Best Run

The explicit bookkeeping winner is

- track1_svm_amplitude_full_closure_split15
It was selected because it achieved:
 - the highest campaign scoped cell-closure score among runs with a real status upgrade;
 - 2 met cells and 3 near cells out of 6 ;
 - the only successful yellow -> green promotion on harmonic 156 across both amplitude surfaces;
 - the strongest scoped final-closure amplitude improvement package.
- Important interpretation:
- this run is the campaign bookkeeping winner;
 - it is **not** the whole scientific result by itself;
 - the scientific output is the merged post-final-closure SVM row now written into the canonical benchmark.

SVM Row Impact

Table-Level Before Vs After

Surface	Before	After
Table 2 amplitude MAE	8 green, 2 yellow, 0 red	9 green, 1 yellow, 0 red
Table 3 amplitude RMSE	7 green, 3 yellow, 0 red	8 green, 2 yellow, 0 red
Table 4 phase MAE	8 green, 1 yellow, 0 red	8 green, 1 yellow, 0 red
Table 5 phase RMSE	8 green, 1 yellow, 0 red	8 green, 1 yellow, 0 red

Closed Or Improved Cells

- Table 2 amplitude MAE : harmonic 156 moved from yellow to green
- Table 3 amplitude RMSE : harmonic 156 moved from yellow to green
- Table 4 phase MAE : harmonic 162 improved numerically but stayed yellow
- Table 5 phase RMSE : harmonic 162 improved numerically but stayed yellow

Residual SVM Yellow Cells

- Table 2 : 40
- Table 3 : 40 , 240
- Table 4 : 162
- Table 5 : 162

There are still no remaining red SVM cells in Tables 2-5 .

Canonical Benchmark Impact

After merging the best final-closure SVM cells into the canonical benchmark:

- Table 2 full-matrix totals moved from 53 / 37 / 10 to 54 / 36 / 10
- Table 3 full-matrix totals moved from 52 / 35 / 13 to 53 / 34 / 13
- Table 4 full-matrix totals stayed at 52 / 37 / 1
- Table 5 full-matrix totals stayed at 43 / 41 / 6

The campaign did not change the canonical harmonic-level Table 6 closure reading. Its impact stays local to the SVM row and the row-level full-matrix surface.

Main Conclusions

The campaign supports four conclusions.

1. The Hardest Remaining SVM Amplitude Cell Was Closed

Amplitude harmonic 156 is no longer yellow on either paper amplitude surface.
That removes the strongest residual amplitude blocker from the SVM row.

2. Phase 162 Improved, But Did Not Close

The best phase run tightened both MAE and RMSE on harmonic 162 .
However, both cells remain yellow, so the phase row is still not fully closed.

3. SVM Is Now A Micro-Repair Problem

The SVM row is no longer a broad repair target.
What remains is a very small residual closure set:

- amplitude 40
- amplitude 240
- phase 162

4. The Best Next Move Is A Tiny Follow-Up Or A Family Shift

At this point, a large new SVM campaign would be inefficient.

The rational options are

- one last tiny SVM micro-closure pass on 40 , 240 , and 162 ; or
- shift the main repair budget to other rows while keeping SVM on a minimal follow-up budget.

Produced Campaign Artifacts

The campaign output folder now contains the required serialized winner artifacts:

- `output/training_campaigns/track1/svm/track1_svm_final_closure_campaign_2026_04_14_20_50_01/campaign_leaderboard.yaml`
- `output/training_campaigns/track1/svm/track1_svm_final_closure_campaign_2026_04_14_20_50_01/campaign_best_run.yaml`
- `output/training_campaigns/track1/svm/track1_svm_final_closure_campaign_2026_04_14_20_50_01/campaign_best_run.md`

The canonical reports updated by this closure are:

- `doc/reports/analysis/rcim_paper_reference/RCIM Paper Reference Benchmark.md`
- `doc/reports/analysis/Training Results Master Summary.md`